

4(a)	RSV / it, is (disease) caused by a pathogen ; (pathogen / RSV, is) transmissible ; A communicable I contagious	2
4(b)	envelope ; A phospholipid bilayer capsid ;	2
4(c)(i)	smooth muscle ;	1
4(c)(ii)	waft / move, mucus, away / up the bronchus / AW ;	1
4(d)(i)	<i>monoclonal antibody</i> : passive and artificial ; <i>vaccination</i> : active and artificial ; A in either order for each treatment	2

Question	Answer	Marks
4(d)(ii)	any four from: 1 (by) injecting (RSV) antigen into mice ; A (other named) small mammal 2 (idea of time) for immune response to occur ; 3 remove / produce, plasma cells / B-lymphocytes / B cells / splenocytes (from the spleen of mice) ; 4 fuse, plasma cells / B-lymphocytes / B cells / splenocytes, and, myeloma / tumour / cancer / AW, cells ; 5 use of fusogen / other named method ; e.g. polyethylene glycol (PEG) 6 screen / select, hybridoma cells ; 7 clone / make many copies of, (selected) hybridoma cells ; 8 AVP ; e.g. production / selection, of one, form of / AW, antibody against RSV antigen humanising antibody / antibody humanisation	4
4(d)(iii)	any one from: long-term immunity / ref. to memory cells / AW ; may contain more than one antigen ; AVP ; e.g. can add adjuvants to obtain a stronger immune response monoclonal antibodies may be destroyed by an immune response	1

Question	Answer	Marks
4(e)(i)	endocytosis occurs / described ; e.g. (phagocyte cell surface) membrane, surrounds / AW, infected cells pseudopodia, surround / form around / AW, infected cell (alveolar) macrophage / phagocyte / (phagocytic) cell, envelops / engulfs, infected cells any other event leading to formation of phagocytic vacuole that occurs after detection of infected body cells ; e.g. <i>before endocytosis</i>: ref. to chemotaxis / chemotactic response phagocyte receptors, bind / attach / join / AW, to viral antigens on cell surface membrane I binding to, macrophage / cell <i>after endocytosis</i>: membrane fusion / (phagocytic), vacuole pinches off / AW A vesicle for vacuole	2
4(e)(ii)	<u>lysosome</u> ;	1

Question	Answer	Marks
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